

File permissions in Linux

Project description

In this project, I examined and modified file permissions in a Linux system to ensure proper authorization. I used Linux commands to check access levels and update permissions to improve system security. This task simulates the real-world responsibilities of a security analyst.

Check file and directory details

In this project, I examined and managed the permissions on the files in the `/home/researcher2/projects` directory for the researcher2 user

1. I navigated to the projects directory using: `cd /home/researcher2/projects`

```
researcher2@86b9e249ba2e:~$ cd projects
researcher2@86b9e249ba2e:~/projects$
```

2. I listed the contents and permissions of the projects directory: `ls -la`

```
researcher2@43565f28df31:~$ cd projects
researcher2@43565f28df31:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 04:18 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 05:29 ..
-rw--w--- 1 researcher2 research_team  46 Apr 14 04:18 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 14 04:18 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Apr 14 04:18 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 14 04:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_t.txt
researcher2@43565f28df31:~/projects$
```

Describe the permissions string

Each file has a 10-character permission string.

Example:

```
-rw-r--r-- 1 researcher2 research_team 46 Apr 14 04:18 project_k.txt
```

Explanation:

- The first character (-) indicates a file (while **d** indicates a directory)
- The next three characters (**rw-**) represent the owner's permissions
- The next three (**r--**) represent the group's permissions
- The last three (**r--**) represent permissions for others

Permissions:

- **r** = read
- **w** = write
- **x** = execute
- **-** = no permission

This means:

- Owner can read and write
- Group can only read
- Others can only read
- characters is a hyphen (-) instead, that indicates that this permission is not granted for other.

The second block of text in the expanded directory listing is the user who owns the file.

The third block of text is the group owner of the file.

Change file permissions

1. Identifying Permission Issues

According to the organization's policy, it does not allow others to have write access to any files.

The following file had incorrect permissions:

- `project_k.txt` (write access for others)

`.project_x.txt`, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file.

- `.project_x.txt` (group had write access but no read access)

The files and directories in the projects directory belong to the researcher2 user. Only researcher2 should be allowed to access the drafts directory and its contents.

- `Drafts` (the group has execute permissions and therefore has access to the drafts directory.)

2. Changing File Permissions

The following files had incorrect write permissions: `project_k.txt` has write permissions for other users. Change the permissions of the file identified in the previous step so that the owner type of other doesn't have write permissions:

```
chmod o-w project_k.txt
```

```

researcher2@43565f28df31:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 04:18 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 05:29 ..
-rw--w---- 1 researcher2 research_team  46 Apr 14 04:18 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 14 04:18 drafts
-rw-r--rw- 1 researcher2 research_team  46 Apr 14 04:18 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 14 04:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_t.txt
researcher2@43565f28df31:~/projects$ chmod o-w project_k.txt
researcher2@43565f28df31:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 04:18 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 05:29 ..
-rw--w---- 1 researcher2 research_team  46 Apr 14 04:18 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 14 04:18 drafts
-rw-r--r-- 1 researcher2 research_team  46 Apr 14 04:18 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 14 04:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_t.txt
researcher2@43565f28df31:~/projects$

```

3. Changing Permissions on a Hidden File

The user and group owner types have incorrect write permissions in the hidden file `.project_x.txt`. To change the permissions of the file `.project_x.txt` so that both the user and the group can read, but not write to, the file used command: `chmod u-w,g-w,g+r .project_x.txt`

```

researcher2@43565f28df31:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 04:18 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 05:29 ..
-rw--w---- 1 researcher2 research_team  46 Apr 14 04:18 .project_x.tx
t
drwx--x--- 2 researcher2 research_team 4096 Apr 14 04:18 drafts
-rw-r--r-- 1 researcher2 research_team  46 Apr 14 04:18 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 14 04:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_t.txt
researcher2@43565f28df31:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@43565f28df31:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 04:18 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 14 05:29 ..
-r--r----- 1 researcher2 research_team  46 Apr 14 04:18 .project_x.tx
t
drwx--x--- 2 researcher2 research_team 4096 Apr 14 04:18 drafts
-rw-r--r-- 1 researcher2 research_team  46 Apr 14 04:18 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Apr 14 04:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 04:18 project_t.txt
researcher2@43565f28df31:~/projects$ █

```

4. Change directory permissions

I removed the execute permission for the group from the drafts directory. The command to complete this step: `chmod g-x drafts`

```
researcher2@86b9e249ba2e:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Apr 14 03:37 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 03:37 project_k.txt
-rw----- 1 researcher2 research_team  46 Apr 14 03:37 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 03:37 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 03:37 project_t.txt
researcher2@86b9e249ba2e:~/projects$ chmod g-x drafts
researcher2@86b9e249ba2e:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Apr 14 03:37 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 03:37 project_k.txt
-rw----- 1 researcher2 research_team  46 Apr 14 03:37 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 03:37 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Apr 14 03:37 project_t.txt
researcher2@86b9e249ba2e:~/projects$
```

This removes group access. For stricter security, permissions could also be set to allow access only for the owner.

Summary

In this project, I reviewed and corrected file permissions to align with organizational security policies. I identified files with excessive write permissions and restricted access using Linux commands. I also handled hidden files and secured a directory to prevent unauthorized access. These actions demonstrate essential skills required for maintaining system security and protecting sensitive data. These changes reduce the risk of unauthorized modification of files.